

chapter-2

$$Q1 \int \frac{(\tan x)^{3/2}}{\cos^2 x} dx$$

$$= \int (\tan x)^{3/2} \sec^2 x dx$$

let,

$$\tan x = z$$

$$\therefore \sec^2 x dx = dz$$

$$= \int z^{3/2} dz$$

$$\Rightarrow \frac{2}{5} z^{5/2} + C = \frac{2}{5} (\tan x)^{5/2} + C$$

$$Q2 \int \sinh^5 x \cosh x dx$$

let,

$$\sinh x = z$$

$$= \int z^5 dz$$

$$\therefore \cosh x dx = dz$$

$$= \frac{1}{6} z^6 + C$$

$$= \frac{1}{6} \sinh^6 x + C$$

$$Q3 \int \frac{e^{a \sin^{-1} x}}{\sqrt{1-x^2}} dx$$

let,

$$\sin^{-1} x = z$$

$$= \int e^{az} dz$$

$$\therefore \frac{dx}{\sqrt{1-x^2}} = dz$$

$$= \frac{1}{a} e^{az} + C = \frac{1}{a} e^{a \sin^{-1} x} + C$$

$$81) \int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$$

$$\text{let } \tan^{-1} x = z$$

$$= \int e^{mz} dz$$

$$\frac{dx}{1+x^2} = dz$$

$$= \frac{1}{m} e^{mz} + C$$

$$= \frac{1}{m} e^{m \tan^{-1} x} + C$$

$$82) \int x e^{2\sqrt{x-1}} dx$$

$$\text{let } x-1 = z$$

$$= \int (z+1) e^{2\sqrt{z}} dz$$

$$x dx = z dz$$

$$= \int (z^2 + 2z + 1) z^{1/2} dz$$

$$= \int (z^{5/2} + 2z^{3/2} + z^{1/2}) dz$$

$$= \frac{2}{7} z^{7/2} + \frac{4}{5} z^{5/2} + \frac{2}{3} z^{3/2} + C$$

$$= \frac{2}{7} (x-1)^{7/2} + \frac{4}{5} (x-1)^{5/2} + \frac{2}{3} (x-1)^{3/2} + C$$

$$= \frac{2}{7} (x-1)^{7/2} + \frac{4}{5} (x-1)^{5/2} + \frac{2}{3} (x-1)^{3/2} + C$$

$$u) \int \frac{\sin(2+3\ln x) dx}{x}$$

let $2+3\ln x = z$

$$= \frac{1}{3} \int \sin z dz$$

$$\frac{dx}{x} = \frac{1}{3} dz$$

$$= \frac{1}{3} \cos z + C$$

$$= \frac{1}{3} \cos(2+3\ln x) + C$$

$$9) \int \frac{(x+1)(x+\ln x)^2}{x} dx$$

let $x+\ln x = z$

$$= \int z^2 dz$$

$$(1 + \frac{1}{x}) dx = dz$$

$$= \frac{1}{3} z^3 + C$$

$$= \frac{1}{3} (x+\ln x)^3 + C$$

$$b) \int \frac{(2x^2+1)(x^2+\ln x) dx}{x}$$

let $x^2+\ln x = z$

$$= \int z dz$$

$$(2x + \frac{1}{x}) dx = dz$$

$$= \frac{1}{2} z^2 + C$$

$$\Rightarrow \frac{2x^2+1}{x} dx = dz$$

$$= \frac{1}{2} (x^2+\ln x)^2 + C$$

$$21) \int \frac{\sinh(x^{-1/2})}{x^{3/2}} dx$$

$$= -2 \int \sinh z dz$$

$$= -2 \cosh z + C$$

$$= -2 \cosh(x^{-1/2}) + C$$

$$\text{let } x^{-1/2} = z$$

$$\frac{dx}{x^{3/2}} = dz (-2)$$

$$22) \int \frac{(1+x)e^x}{\cos^2(xe^x)} dx$$

$$= \int \frac{dz}{\cos^2 z}$$

$$= \int \sec^2 z dz$$

$$= \tan z + C = \tan(xe^x) + C$$

$$\text{let } xe^x = z$$

$$(xe^x + e^x) dx = dz$$

$$\therefore e^x(1+x) dx = dz$$

$$23) \int x 7^{x^2} dx$$

$$= \int \frac{z}{2 \ln 7} dz$$

$$= \frac{1}{2 \ln 7} \int dz$$

$$= \frac{1}{2 \ln 7} z + C$$

$$= \frac{1}{2 \ln 7} 7^{x^2} + C$$

$$\text{let } x^2 = z$$

$$2x \ln 7 = \ln z$$

$$2x \ln 7 dx = \frac{dz}{z}$$

$$\therefore z dx = \frac{dz}{2 \ln 7}$$

22) $\int \frac{dx}{e^x + 1}$

$= \int \frac{1 + e^x - e^x}{e^x + 1} dx$

$= \int dx - \int \frac{e^x}{e^x + 1} dx$

$= x - \int \frac{dz}{z}$

$= x - \ln z + C = x - \ln(e^x + 1) + C$

let, $e^x + 1 = z$

$e^x dx = dz$

26) $\int \frac{\sin x dx}{(1 - \cos x)^2}$

$= \int \frac{dz}{z^2}$

$= -\frac{1}{z} + C$

$= -\frac{1}{1 - \cos x} + C$

let, $1 - \cos x = z$

$\sin x dx = dz$

$\sin x dx = dz$

28) $\int \frac{\sqrt{x} + \ln x}{x} dx$

$= \int \frac{\sqrt{x}}{x} dx + \int \frac{\ln x}{x} dx$

$= 2\sqrt{x} + \int z dz$

$= 2\sqrt{x} + \frac{1}{2} z^2 + C$

$= 2\sqrt{x} + \frac{1}{2} (\ln x)^2 + C$

let, $\ln x = z$

$\frac{dx}{x} = dz$

$$25) \int x \sqrt{1+x} dx$$

$$= \int (z+1) \sqrt{z} dz$$

$$= \int (z^{3/2} + z^{1/2}) dz$$

$$= \frac{2}{5} z^{5/2} + \frac{2}{3} z^{3/2} + C$$

$$= \frac{2}{5} (x+1)^{5/2} + \frac{2}{3} (x+1)^{3/2} + C$$

Let $1+x = z$
 $dx = dz$

Let $x = z-1$

$$26) \int x \sqrt[3]{(1-x^2)^5} dx$$

$$= -\frac{1}{2} \int \sqrt[3]{z^5} dz$$

$$= -\frac{1}{2} \int z^{5/3} dz$$

$$= -\frac{1}{2} \times \frac{3}{8} z^{8/3} + C$$

$$= -\frac{3}{16} (1-x^2)^{8/3} + C$$

Let $1-x^2 = z$

$-2x dx = dz$

$x dx = -\frac{1}{2} dz$

$$\underline{29)} \int \frac{x^2 dx}{\sqrt{x+1}}$$

$$= \int \frac{z^2 - 2z + 1}{\sqrt{z}} dz$$

$$= \int \left(z^{3/2} - 2z^{1/2} + \frac{1}{\sqrt{z}} \right) dz$$

$$= \frac{2}{5} z^{5/2} - \frac{4}{3} z^{3/2} + 2\sqrt{z} + C$$

$$= \frac{2}{5} (x+1)^{5/2} - \frac{4}{3} (x+1)^{3/2} + 2\sqrt{x+1} + C$$

$$\underline{26)} \int \frac{x^2}{\sqrt{4+x}} dx$$

$$= \int \frac{z^2 - 8z + 16}{\sqrt{z}} dz$$

$$= \int \left(z^{3/2} - 8z^{1/2} + \frac{16}{\sqrt{z}} \right) dz$$

$$= \frac{2}{5} z^{5/2} - \frac{16}{3} z^{3/2} + 16 \times 2\sqrt{z} + C$$

$$= \frac{2}{5} (4+x)^{5/2} - \frac{16}{3} (4+x)^{3/2} + 32\sqrt{4+x} + C$$

$$\text{let } x+1 = z$$

$$x+1 = z$$

$$dx = dz$$

$$x^2 = z^2 - 2z + 1$$

$$\text{let } 4+x = z$$

$$4+x = z$$

$$dx = dz$$

$$x^2 = z^2 - 8z + 16$$

$$20) \int \frac{x^3}{\sqrt{2-x^2}} dx$$

let

$$2-x^2 = z$$

$$x dx = -\frac{1}{2} dz$$

$$x^2 = 2-z$$

$$= -\frac{1}{2} \int \frac{2-z}{\sqrt{z}} dz$$

$$= -\frac{1}{2} \int \left(\frac{2}{\sqrt{z}} - \sqrt{z} \right) dz$$

$$= -\frac{1}{2} \times 2\sqrt{z} + \frac{1}{2} \times \frac{2}{3} z^{3/2} + C$$

$$= \frac{1}{3} (2-x^2)^{3/2} - \sqrt{2-x^2} + C$$

$$20) \int \frac{x^3}{\sqrt{x-1}} dx$$

$$x-1 = z$$

$$dx = dz$$

$$x = z+1$$

$$= \int \frac{z^3 + 3z^2 + 3z + 1}{\sqrt{z}} dz$$

$$x^3 = z^3 + 3z^2 + 3z + 1$$

$$= \int \left(z^{5/2} + 3z^{3/2} + 3z^{1/2} + \frac{1}{\sqrt{z}} \right) dz$$

$$= \frac{2}{7} z^{7/2} + \frac{6}{5} z^{5/2} + 3 \cdot \frac{2}{3} z^{3/2} + 2\sqrt{z} + C$$

$$= \frac{2}{7} (x-1)^{7/2} + \frac{6}{5} (x-1)^{5/2} + 2(x-1)^{3/2} + 2\sqrt{x-1} + C$$

$$\underline{221} \quad \int x^3 \sqrt{5-x^3} dx$$

$$= \int x^3 \sqrt{5-x^3} \cdot x^2 dx$$

$$= -\frac{4}{3} \int (5-z) \sqrt{z} dz$$

$$= -\frac{4}{3} \int (5z^{1/2} - z^{3/2}) dz$$

$$= -\frac{4}{3} \left[5 \times \frac{2}{3} z^{3/2} - \frac{2}{5} z^{5/2} \right] + C$$

$$= -\frac{40}{9} (5-x^3)^{3/2} + \frac{2}{15} (5-x^3)^{5/2} + C$$

$$\underline{222} \quad \int \frac{dx}{x \sqrt{x^2-1}}$$

$$= \int \frac{\sec \theta \tan \theta d\theta}{\sec \theta \sqrt{\sec^2 \theta - 1}}$$

$$= \int \frac{\tan \theta d\theta}{\tan \theta}$$

$$= \int d\theta$$

$$= \theta + C$$

$$= \sec^{-1} x + C$$

$$\text{let } 5-x^3 = z$$

$$5-x^3 = z$$

$$\Rightarrow -3x^2 dx = dz$$

$$\Rightarrow x^2 dx = -\frac{1}{3} dz$$

$$x^3 = 5-z$$

$$\text{let } x = \sec \theta$$

$$x = \sec \theta$$

$$dx = \sec \theta \tan \theta d\theta$$

$$\theta = \sec^{-1} x$$

$$\underline{26)} \int \frac{dx}{x\sqrt{x^4-1}} \quad \text{let } x^2 = \sec \theta$$

$$= \int \frac{x dx}{x^2 \sqrt{x^4-1}} \quad \text{let } x^2 = \sec \theta \Rightarrow x = \sec^{1/2} \theta$$

$$= \frac{1}{2} \int \frac{\sec \theta \tan \theta d\theta}{\sec \theta \sqrt{\sec^2 \theta - 1}}$$

do

$$2x dx = \sec \theta \tan \theta d\theta$$

$$= \frac{1}{2} \int d\theta$$

$$= \frac{1}{2} \theta + C$$

$$= \frac{1}{2} \sec^{-1} x^2 + C$$

$$\underline{28)} \int \frac{\sqrt{\tan x}}{\sin x \cdot \cos x} dx$$

$$= \int \frac{\sqrt{\tan x}}{\frac{\sin x}{\cos x} \cdot \cos x} dx$$

$$= \int \frac{\sec^2 x}{\sqrt{\tan x}} dx$$

$$\text{let } \tan x = z$$

$$\sec^2 x dx = dz$$

$$= \int \frac{1}{\sqrt{z}} dz$$

$$= 2\sqrt{z} + C$$

$$= 2\sqrt{\tan x} + C$$

$$\underline{24)} \int \frac{\sin 2x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2} dx$$

$$= \int \frac{2 \sin x \cos x}{(a^2 \sin^2 x + b^2 \cos^2 x)^2} dx$$

$$= \frac{1}{a^2 - b^2} \int \frac{dz}{z^2}$$

$$= \frac{1}{a^2 - b^2} \left(-\frac{1}{z} \right) + C$$

$$= \frac{1}{b^2 - a^2} \frac{1}{(a^2 \sin^2 x + b^2 \cos^2 x)} + C$$

$$\underline{25)} \int \frac{\sin 2x}{a^2 \sin^2 x + b^2 \cos^2 x} dx$$

$$= \frac{1}{a^2 - b^2} \int \frac{dz}{z}$$

$$= \frac{1}{a^2 - b^2} \ln z + C$$

$$= \frac{1}{a^2 - b^2} \ln (a^2 \sin^2 x + b^2 \cos^2 x) + C$$

$$29) \int \frac{\sin x \cos x}{\cos^4 x + \sin^4 x} dx$$

$$= \int \frac{\frac{\sin x}{\cos x} \sec^2 x}{1 + \tan^4 x} dx$$

$$= \int \frac{\tan x \sec^2 x}{1 + \tan^4 x} dx \quad \tan^2 x = z$$

$$= \frac{1}{2} \int \frac{dz}{1+z^2}$$

$$= \frac{1}{2} \tan^{-1} z + C$$

$$= \frac{1}{2} \tan^{-1} \tan^2 x + C$$

$$30) \int \frac{dx}{(x^2+1)^2}$$

$$= \int \frac{\sec^2 \theta d\theta}{(\tan^2 \theta + 1)^2}$$

$$= \int \frac{\sec^2 \theta d\theta}{\sec^4 \theta}$$

$$= \int \cos^2 \theta d\theta$$

$$= \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta + C$$

$$\text{let } x = \tan \theta \quad dx = \sec^2 \theta d\theta$$

$$= \frac{1}{2} \tan^{-1} x + \frac{1}{4} \frac{2 \tan \theta}{1 + \tan^2 \theta} + C$$

$$= \frac{1}{2} \tan^{-1} x + \frac{1}{2} \frac{2x}{1+x^2} + C$$

$$2.2) \int \frac{dx}{(1+x^2)^{3/2}}$$

$$= \int \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^{3/2}}$$

$$= \int \frac{\sec^2 \theta d\theta}{\sec^3 \theta}$$

$$= \int \cos \theta d\theta$$

$$= \sin \theta + c$$

$$= \sin \tan^{-1} x + c$$

$$= \frac{x}{\sqrt{1+x^2}} + c$$

$$\frac{x}{\sqrt{1+x^2}} + c$$

$$100) \int \frac{\sin 2x}{(a+b \cos x)^2} dx$$

$$= \int \frac{2 \sin x \cdot \cos x}{(a+b \cos x)^2} dx$$

$$= -\frac{2}{b^2} \int \frac{(z-a) dz}{z^2}$$

$$= -\frac{2}{b^2} \int \left(\frac{1}{z} - \frac{a}{z^2} \right) dz$$

$$x^2 + 1 = \tan^2 \theta$$

$$dx = \sec^2 \theta d\theta$$

$$\tan \theta = \tan^{-1} x$$



$$\sin \theta = \frac{x}{\sqrt{1+x^2}}$$

$$\cos \theta = \frac{1}{\sqrt{1+x^2}}$$

$$\sec \theta = \sqrt{1+x^2}$$

$$\tan \theta = x$$

$$\sec^2 \theta = 1 + \tan^2 \theta = 1 + x^2$$

$$\sec \theta = \sqrt{1+x^2}$$

$$\tan \theta = x$$

$$\sec^2 \theta = 1 + \tan^2 \theta = 1 + x^2$$

$$\sec \theta = \sqrt{1+x^2}$$

$$\tan \theta = x$$

$$= -\frac{2}{b^2} \left[\ln z - \frac{a}{z} \right] + c$$

$$= -\frac{2}{b^2} \ln(a + b \cos x) + \frac{2a}{b^2} \times \frac{1}{a + b \cos x} + c$$

Q2) $\int \sqrt{1 + \sec x} dx$

$$= \int \sqrt{1 + \frac{1}{\cos x}} dx$$

$$= \int \sqrt{\frac{1 + \cos x}{\cos x}} dx$$

$$= \int \frac{\sqrt{2 \cos^2 x/2}}{\sqrt{1 - 2 \sin^2 x/2}} dx$$

Let,

$$\sqrt{2} \sin x/2 = z$$

$$= \int \frac{\sqrt{2} \cos x/2}{\sqrt{1 - 2 \sin^2 x/2}} dx \quad \sqrt{2} \cos x/2 dx = 2dz$$

$$= \int \frac{2 dz}{\sqrt{1 - z^2}}$$

$$= 2 \sin^{-1} z + c$$

$$= 2 \sin^{-1} (\sqrt{2} \sin x/2) + c$$

$$= 2 \sin^{-1} (\sqrt{1 - \cos x}) + c$$

$$(Q2) \int \cos 2 \cot^{-1} \sqrt{\frac{1-x}{1+x}} dx$$

Let.

$$x = \cos \theta$$

$$\theta = \cos^{-1} x$$

$$dx = -\sin \theta d\theta$$

$$= \int \cos 2 \cot^{-1} \frac{\sqrt{1-\cos \theta}}{\sqrt{1+\cos \theta}} (-\sin \theta) d\theta$$

$$= - \int \cos 2 \cot^{-1} \frac{\sqrt{2} \sin \theta / 2}{\sqrt{2} \cos \theta / 2} \cdot \sin \theta d\theta$$

$$= - \int \cos 2 \cot^{-1} \tan \theta / 2 \cdot \sin \theta d\theta$$

$$= - \int \cos 2 \cot^{-1} \cot \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \cdot \sin \theta d\theta$$

$$= - \int \cos 2 \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \cdot \sin \theta d\theta$$

$$= \int -\cos(\pi - \theta) \cdot \sin \theta d\theta$$

$$= \frac{1}{2} \int 2 \cos \theta \cdot \sin \theta d\theta$$

$$= \frac{1}{2} \int \sin 2\theta d\theta$$

$$= -\frac{1}{4} \cos 2\theta + C$$

$$= -\frac{1}{4} (2\cos^2 \theta - 1) + C$$

$$= -\frac{1}{4} (2x^2 - 1) + C$$

$$\underline{Q81} \int \frac{\sqrt{x} dx}{\sqrt{a^3 - x^3}}$$

$$= \frac{2}{3} \int \frac{dz}{\sqrt{a^3 - z^2}}$$

$$= \frac{2}{3} \sin^{-1} \frac{z}{a^{3/2}} + C$$

$$= \frac{2}{3} \sin^{-1} \frac{\sqrt{x^3}}{\sqrt{a^3}} + C$$

$$\underline{Q82} \int \frac{dx}{x^{1/2} - x^{1/4}}$$

$$= \int \frac{4z^3 dz}{z^2 - z}$$

$$= 4 \int \frac{z^2 dz}{z-1}$$

$$= 4 \int \frac{z^2 - 1 + 1}{z-1} dz$$

$$= 4 \int (z+1) dz + 4 \int \frac{dz}{z-1}$$

$$= 4 \frac{z^2}{2} + 4z + 4 \ln |z-1| + C$$

$$= 2z^2 + 4z + 4 \ln |z-1| + C$$

$$= 2x^{1/2} + 4x^{1/4} + 4 \ln |x^{1/4} - 1| + C$$

$$\text{Let } x^{3/2} = z$$

$$\frac{3}{2} x^{1/2} dx = dz$$

$$\sqrt{x} dx = \frac{2}{3} dz$$

again,

$$x^3 = z^2$$

$$\text{Q6)} \int \sqrt{\frac{x}{a-x}} dx$$

Let $x = a \sin^2 \theta$
 $dx = 2a \sin \theta \cos \theta d\theta$

$$= \int \frac{\sqrt{a \sin^2 \theta}}{\sqrt{a - a \sin^2 \theta}} \cdot 2a \sin \theta \cos \theta d\theta$$

$$= 2a \int \frac{\sin \theta}{\cos \theta} \cdot \sin \theta \cos \theta d\theta$$

$$= 2a \int \sin^2 \theta d\theta$$

$$= 2a \int (1 - \cos 2\theta) d\theta$$

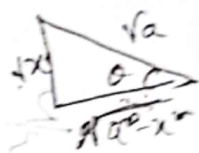
$$= 2a\theta - \frac{a}{2} \sin 2\theta + C$$

$$= a \sin^{-1} \sqrt{\frac{x}{a}} - \frac{a}{2} 2 \sin \theta \cos \theta + C$$

$$= a \sin^{-1} \sqrt{\frac{x}{a}} - a \sin \theta \cos \theta + C$$

$$= a \sin^{-1} \sqrt{\frac{x}{a}} - a \frac{\sqrt{x}}{\sqrt{a}} \cdot \frac{\sqrt{a-x}}{\sqrt{a}} + C$$

$$= a \sin^{-1} \sqrt{\frac{x}{a}} - \sqrt{x(a-x)} + C$$



Q9) $\int \frac{x dx}{\sqrt{x+1}}$

$= 2 \int \frac{(z^2-1)z dz}{z}$

$= 2 \int (z^2-1) dz$

$= 2 \cdot \frac{1}{3} z^3 - 2z + C$

$= \frac{2}{3} (\sqrt{x+1})^3 - 2(\sqrt{x+1}) + C$

Answer, $\int \frac{x dx}{\sqrt{x+1}}$

$= \int \frac{x+1-1}{\sqrt{x+1}} dx$

$= \int \sqrt{x+1} dx - \int \frac{1}{\sqrt{x+1}} dx$

$= \frac{2}{3} (\sqrt{x+1})^3 - 2\sqrt{x+1} + C$

Let, $x+1 = z^2$

$dx = 2z dz$

$x = z^2 - 1$

Q. 61

$$\int \frac{x^{1/2} dx}{(1+x)^{5/2}}$$

$$= \int \frac{-x^2 dz}{x^{1/2} z^{5/2} x^{5/2}}$$

$$= - \int \frac{dz}{x \cdot z^{5/2}}$$

$$= - \int (z-1) z^{-5/2} dz$$

$$= - \int (z^{-3/2} - z^{-5/2}) dz$$

$$= - \left(2z^{-1/2} - \frac{2}{3} z^{-3/2} \right) + C$$

$$= - \frac{2}{\sqrt{z}} + \frac{2}{3\sqrt{z^3}} + C$$

$$= - \frac{2\sqrt{x}}{\sqrt{1+x}} + \frac{2}{3} x \frac{1}{\sqrt{(1+x)^3}} + C$$

Let

$$1+x = z^2$$

$$x = \frac{1}{z^2} - 1$$

$$z = \frac{1}{\sqrt{x}} + 1$$

$$dz = -\frac{1}{2x^{3/2}} dx$$

$$dx = -2x^2 dz$$

Q2) $\int \frac{1 dx}{3\sqrt{x} \cdot 3\sqrt{(1+x)^5}}$ let $1+x = z$
 $1 = x(z-1)$

$= \int \frac{1 dx}{x^{1/3} \cdot (1+x)^{5/3}}$ $\frac{1}{dx} \frac{dz}{dz} = -\frac{1}{(z-1)^2} dz$

$= \int \frac{-\frac{dz}{(z-1)^2}}{x^{1/3} \cdot (z)^{5/3}}$

$= \int \frac{-\frac{dz}{(z-1)^2}}{x^2 \cdot z^{5/3}}$

$= - \int \frac{dz}{z^{5/3}}$

$= - \frac{3}{2} z^{-2/3} + C$

$= - \frac{3}{2} \left(\frac{1+x}{x}\right)^{-2/3} + C$

$= - \frac{3}{2} \left(\frac{x}{1+x}\right)^{2/3} + C$

$$80) \int \frac{x^2 + 1}{x^4 + 1} dx$$

$$x - \frac{1}{x} = z$$

$$\int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$\left(1 + \frac{1}{x^2}\right) dx = dz$$

$$\frac{1}{1-x} = x^2 + \frac{1}{x^2}$$

$$= \int \frac{1 + \frac{1}{x^2}}{\left(1 - \frac{1}{x}\right)^2 + 2} dx$$

$$= \int \frac{dz}{z^2 + (\sqrt{2})^2}$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{z}{\sqrt{2}} + C$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{x - \frac{1}{x}}{\sqrt{2}} + C$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \frac{x^2 - 1}{\sqrt{2} x} + C$$